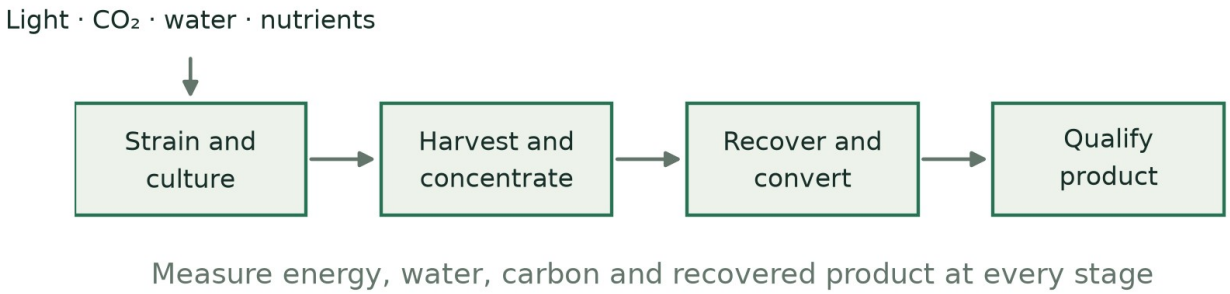
Algae fuels and materials

STUDENT STUDY GUIDE

Biochemical Engineering · Years 2 and 3 · JUSTLAB

English review edition · 8 November 2026

Algae can supply carbon for fuels and materials, and they can serve as hosts for useful enzymes. Whether an algal route improves sustainability depends on the complete process: how the organism grows, what is recovered, how the product is used and what happens at the end of its life. This guide connects those decisions to chemistry, biology and engineering.



The engineering question extends from the organism to a qualified product. Original teaching schematic.

What you should be able to do

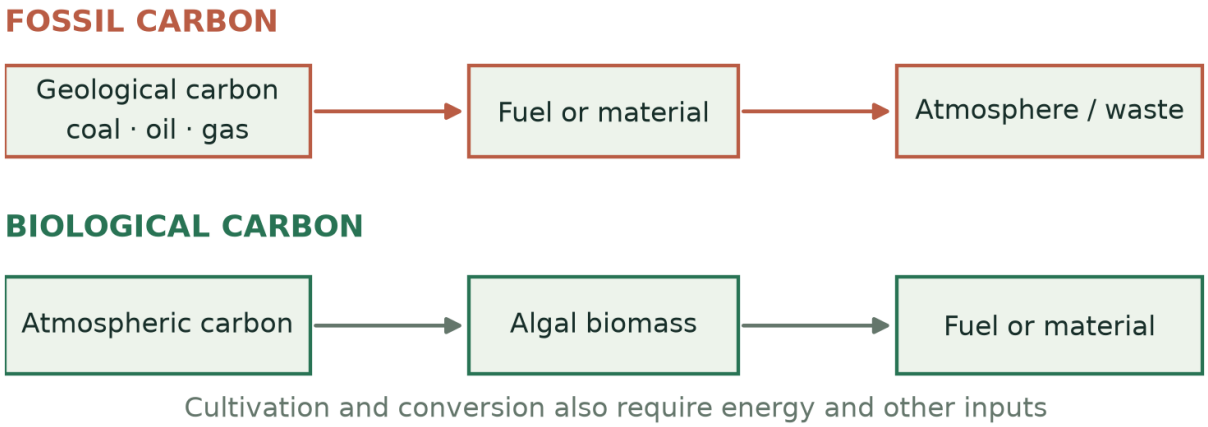
Calculate a weekly commuting estimate and explain its assumptions; distinguish total energy from electricity and CO_2 from CO_2e; explain why batteries are useful for many transport services and challenging for some aircraft and ships; distinguish biodiesel from aviation hydrocarbons; interpret the algae research without confusing cellular content with process productivity.

Read the core explanations first, then attempt the practice problems before opening the answers. Sources are identified by reference codes such as [R10], with links at the end. Illustrative calculations are labelled separately from reported experimental results.

This edition accompanies a one-hour class with an initial commuting activity and a final Algae Farm activity. It is the Stage 2 review version; a Portuguese edition and updated presentation follow in Stage 3.

Carbon as an energy source and a material feedstock

Petroleum has two roles in this class. Its molecules store chemical energy, and their carbon skeletons supply feedstocks for materials. Replacing petroleum therefore requires more than replacing a power station: some applications need electricity, while others need particular molecules and material properties.



Carbon origin matters, but energy use, conversion losses and end of life also enter the assessment. The biological route is not automatically carbon neutral.

Why the carbon cycle matters

Combustion transfers fuel carbon into CO_2. Oxidizing one mole of carbon produces one mole of CO_2, so 12 g of carbon corresponds to 44 g of CO_2. Fossil fuel use transfers carbon from geological stores into the active atmosphere and biosphere. Carbon recently taken up by algae can circulate on a shorter timescale, but emissions from cultivation and conversion still need accounting.

$$m_{CO_2} = m_C \times \frac{44}{12}$$

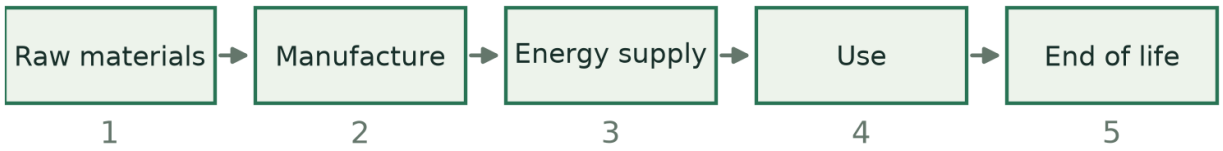
For an illustrative 1 kg of dry biomass containing 50% carbon, the carbon corresponds to 0.50 × 44/12 = 1.83 kg CO_2. The 50% carbon content is an assumed teaching input. This is a carbon balance, not an avoided-emissions result. The biomass may later release that carbon through fuel combustion or biodegradation.

From biology to a climate claim

Photosynthetic uptake does not by itself establish permanent removal. To claim an advantage over a fossil product, define the alternative displaced, count the process inputs and compare equivalent services. Land-use effects, hydrogen production and coproduct treatment can materially affect fuel life-cycle results. [R18, R22]

How to compare environmental footprints

A life-cycle assessment follows the activities needed to provide a defined service. Its functional unit states what is being compared: one passenger-kilometre, one megajoule of fuel, or a square metre of coated fabric delivering a specified performance and lifetime. A comparison per kilogram of material can be misleading when the materials perform differently. [R08–R10, R18]



A possible life-cycle boundary. A real study must state which stages and infrastructure are included.

Keep the units and boundaries together

A vehicle-kilometre measures vehicle movement. A passenger-kilometre measures transport of one person over one kilometre. If a vehicle travels 10 km with four people, it provides 40 passenger-km. Allocation across passengers can reduce the footprint assigned to each person without changing the vehicle trip.

CO_2 describes one gas. CO_2e expresses effects using a selected climate metric. Its value depends on the accounting method and time horizon. The flight scenario used in the class also includes contrail effects, whereas a combustion-only CO_2 calculation would exclude them. A numerical comparison is meaningful only when those choices are clear. [R32–R35]

A bicycle illustrates boundary choices

The ECF/TNO example assumes eight years at 2,400 km/year, giving 19,200 km. It assigns about 5 g CO_2e/km to manufacture and maintenance. A separate dietary assumption brings its cycling scenario to about 21 g/km. Infrastructure and disposal are outside that comparison. [R08]

The live dashboard instead uses a frozen ADEME reference with no extra dietary emissions. Its bicycle value is 0.17 g/km. These are different models, not quantities to add together. A useful sensitivity exercise is to calculate the journey under each complete boundary and explain why the answers differ. [R05]

Before accepting a footprint number, ask which service, location, year, lifetime, occupancy and excluded processes it represents.

Entering and interpreting the commuting poll

Describe a typical week using journeys of ida or volta. Going to campus and returning home counts as two journeys. Someone can travel out by bus and return by bicycle. The current form accepts up to 14 one-way journeys and a single approximate one-way distance. No home address is required.

$$\text{Weekly distance} = \text{one-way distance} \times \text{journeys}$$

$$\text{Weekly estimate} = \sum_{m} (\text{distance}_m \times \text{factor}_m)$$

The selected factors below are teaching references for France, not measured averages for the Brazilian fleet. Manufacturing and fuel-chain use are included; infrastructure and additional food are excluded. Values are rounded here for readability; the application calculates with the frozen constants. [R05–R07]

Mode	g CO_2e per passenger km	Main assumption
Car alone	169.708	Reference gasoline car
Bus	122.420	Mean occupancy already included
Uber or taxi	169.708	Car proxy; empty running excluded
Mechanical bicycle	0.170	Selected database manufacture factor
Car with 2 total occupants	84.854	Vehicle impact divided by 2
Car with 3 total occupants	56.569	Vehicle impact divided by 3
Car with 4 total occupants	42.427	Vehicle impact divided by 4
Car with 5 total occupants	33.942	Vehicle impact divided by 5
Motorcycle up to 250 cm^3	87.200	Selected reference category
Walking	0	No extra food in this boundary

“Carona +1” means two occupants in total, including the driver. The bus factor is already per passenger-km, so dividing it again by the number on the bus would count occupancy twice. The Uber proxy counts one paying passenger; the driver is not a second passenger for this allocation.

“Other/combined” has no defined factor and remains unestimated. It must not become zero emissions. The shared distance also limits precision when outward and return routes differ, or when a journey contains several modes. Report those limitations rather than presenting the result as a personal measurement.

A worked weekly footprint and class total

The following is a synthetic example, not real student data. Ana reports 14 journeys of 8 km: seven by bus and seven by bicycle. Each mode therefore covers 56 passenger-km. Use factors in kilograms by dividing the displayed grams by 1,000.

Quantity	Calculation	Result
Weekly passenger distance	8 × 14	112 km
Bus estimate	56 × 0.12242	6.85552 kg CO_2e
Bicycle estimate	56 × 0.00017	0.00952 kg CO_2e
Ana weekly total	6.85552 + 0.00952	6.86504 kg CO_2e
Average per journey	6.86504 ÷ 14	0.49036 kg CO_2e

From two people to a class

Bruno reports ten 8-km journeys by car. His distance is 80 passenger-km and the estimate is 80 × 0.1697080247 = 13.57664 kg CO_2e/week. For this two-person class, sum the individual distances and emissions.

$$D_{\text{class}} = 112 + 80 = 192 \text{ passenger-km/week}$$

$$E_{\text{class}} = 6.86504 + 13.57664 = 20.44168 \text{ kg } \text{CO}_2\text{e/week}$$

$$\text{Mean} = \frac{20.44168}{2} = 10.22084 \text{ kg } \text{CO}_2\text{e/person/week}$$

These are three different outputs: a personal total, a class total and a mean per respondent. The application includes zero-journey respondents in that mean. A person reporting a zero-journey week lowers the mean but does not reduce the emissions already reported by other people.

When the total is partial

If someone reports an unsupported mode, the dashboard can sum the covered portions, but the total and mean remain partial. Coverage should be calculated from estimated distance divided by all reported distance. Reporting 90% distance coverage is more informative than saying that 90% of students had at least one supported mode.

Distances summed across people are passenger-kilometres. They are not a count of unique vehicle-kilometres. For class trend comparisons, retain the scenario version, response count and the same question wording; changes in participation and assumptions can change the total without a change in individual behaviour. [R05]

Comparing the class distance with flying

The class asks a counterfactual question: what would the climate estimate be if the same passenger-distance were provided by a selected flight profile? This keeps distance fixed while changing the transport reference. It does not invent a literal flight whose route equals the sum of all commutes.



Synthetic example for Ana using ADEME factors. The short-flight profile includes contrail effects; values are kg CO_2e. [R05, R32]

ADEME flight profile	Source route band	kg CO_2e per passenger km
Short	300 to 1,000 km	0.225
Medium	1,000 to 2,000 km	0.185
Long	More than 5,000 km	0.178

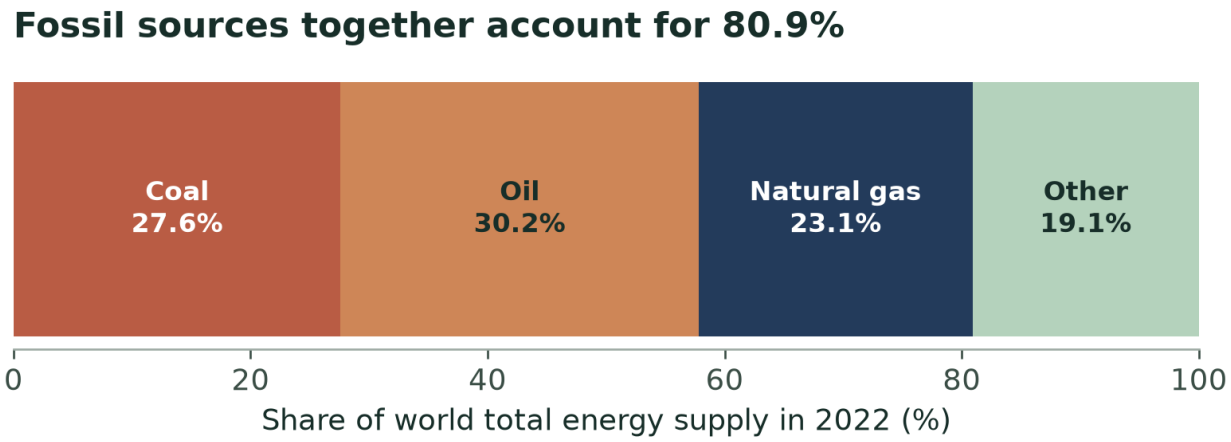
Ana's 112 passenger-km × 0.225 gives 25.2 kg CO_2e. Dividing by her 6.86504-kg commuting estimate gives a ratio of 3.67. For the synthetic two-person class, 192 × 0.225 = 43.2 kg, or 2.11 times the 20.44168-kg commuting total. Ratios depend on the class's reported modes. [R32–R34]

The three bands are representative scenarios, not a continuous route model. A comparison with an actual flight needs its route, cabin class, number of passengers and whether the journey is one-way or return. A CO_2-only route calculator and this CO_2e scenario do not have identical boundaries.

Do not force a preselected conclusion that a flight must be a fixed number of times worse. Report the assumptions and let the calculated result determine the comparison. A ratio is undefined when the commuting denominator is zero, and it is misleading when that denominator omits unknown-mode distance.

The scale and composition of energy use

The historical graph in the presentation traces global fossil primary-energy consumption from 1800 to 2023. It shows coal, oil and gas in TWh. It provides a quantitative view of industrial-era growth, not a measured record covering thousands of years of all human energy use. The underlying chart credits the Energy Institute and Smil. [R03]



Historical teaching snapshot from supplied Lecture 1, PDF p. 9. The exact upstream dataset vintage remains unpinned. The corrected fossil sum is 80.9%, not 81.7%. [R01]

Energy and electricity answer different questions

Total energy supply includes fuels that may be burned directly as well as energy entering electricity generation. Electricity is an energy carrier: it can come from several primary sources. A high renewable share in electricity does not imply that the same share of transport fuels, industrial heat and total energy is renewable.

In the supplied 2022 world snapshot, coal, oil and natural gas account for 27.6%, 30.2% and 23.1%, respectively. Their sum is 80.9%. “Other” includes several sources with different characteristics; it cannot be renamed “renewables” or “zero carbon.” Nuclear is generally classified as nonrenewable but can have low life-cycle GHG emissions. [R01, R02]

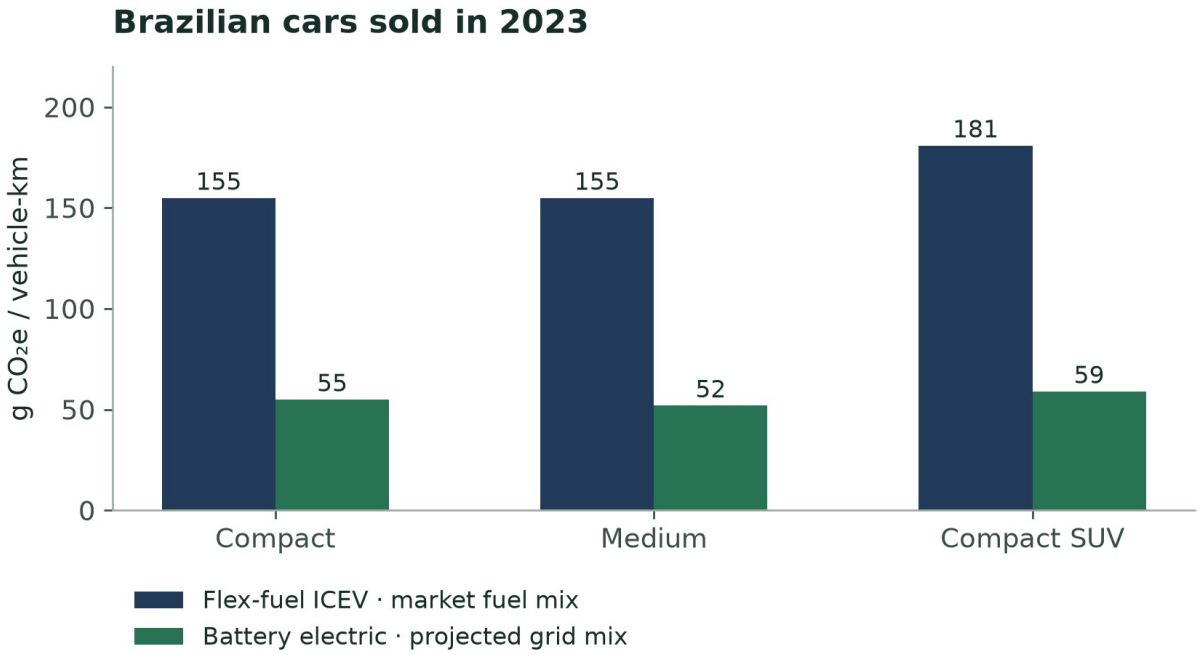
A Brazilian complement

EPE’s BEN 2026 reports that renewables accounted for 86.8% of Brazil’s domestic electricity supply in 2025, including net imports. This is a different geography, year and denominator from the world total-energy chart. The earlier 88.2% electricity figure belongs to 2024. Neither percentage is itself a life-cycle emission factor. [R11]

Historical energy datasets also change methodology. OWID revised its primary-energy method in August 2026. Preserve the year, definition and source version when updating a graph rather than silently replacing one percentage with another. [R02]

Transport choices and electric vehicles

In the presentation's historical 2018 breakdown, road passenger transport accounts for 45.1% of transport CO_2 and road freight for 29.4%, giving 74.5% together. Aviation is 11.6% and shipping 10.6%. These are shares of transport CO_2, not all global greenhouse gases. Rail plus other transport gives 3.2%; rounding makes the total 99.9%. [R04]



ICCT model for vehicles sold in Brazil in 2023. Central model estimates, including manufacturing and energy supply; lifetime 22 years and 288,000 km. [R10]

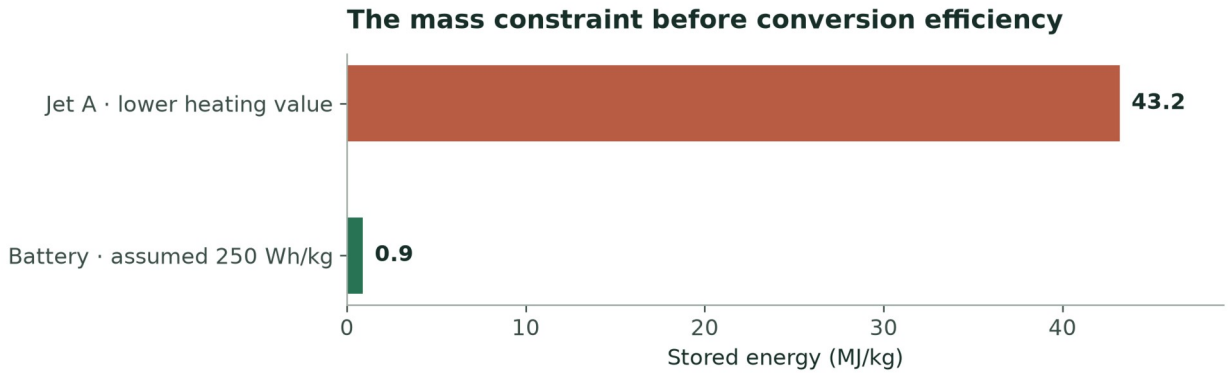
For the compact segment, 155 versus 55 g CO_2e/km means approximately 65% lower lifetime emissions for the BEV in that matched scenario. Vehicle size still matters: the compact SUV estimates differ from those for smaller cars. The model's fuel and grid mixes extend over 2023–2044; this is a lifetime projection, not a measurement of every car currently on the road. [R10]

The EU example in the current presentation is another scenario: 235 g CO_2e/km for a medium petrol car versus 63 for a BEV on the projected EU grid and 52 with renewable electricity, for cars sold in 2025. These cases imply reductions of about 73% and 78%. Keep them labelled EU rather than transferring the numbers to Brazil. [R09]

Electrification can lower impacts while reducing car dependence can change how much vehicle travel is needed. Walking, cycling, accessible public transport and shared trips address different services and constraints. A fair class discussion considers safety, distance, accessibility and occupancy as well as vehicle technology.

Why some transport services are harder to electrify

An energy carrier must fit the vehicle's mass, volume, power and operating pattern. Batteries are attractive where charging and trip length suit them. Long-range aviation is especially sensitive to the mass that must be carried. Ocean shipping presents different constraints, and some short routes can use batteries. Rail can obtain electricity from overhead wires rather than carrying its entire energy supply.



Jet A lower heating value 43.2 MJ/kg; battery teaching assumption 250 Wh/kg = 0.9 MJ/kg. Stored energy is not useful propulsion energy. [R13, R14]

$$250\ \mathrm{Wh/kg} \times 3.6\ \mathrm{kJ/Wh} = 900\ \mathrm{kJ/kg} = 0.9\ \mathrm{MJ/kg}$$

$$\text{Stored energy ratio} = \frac{43.2}{0.9} = 48$$

The 48-fold ratio does not mean a 48-fold difference in range. Electric motors and turbines convert stored energy with different efficiencies. Aircraft fuel mass decreases during flight, whereas discharged batteries still weigh almost the same. Pack structure, reserves, cooling and the vehicle's aerodynamics must enter an actual design.

Efficiency changes the comparison

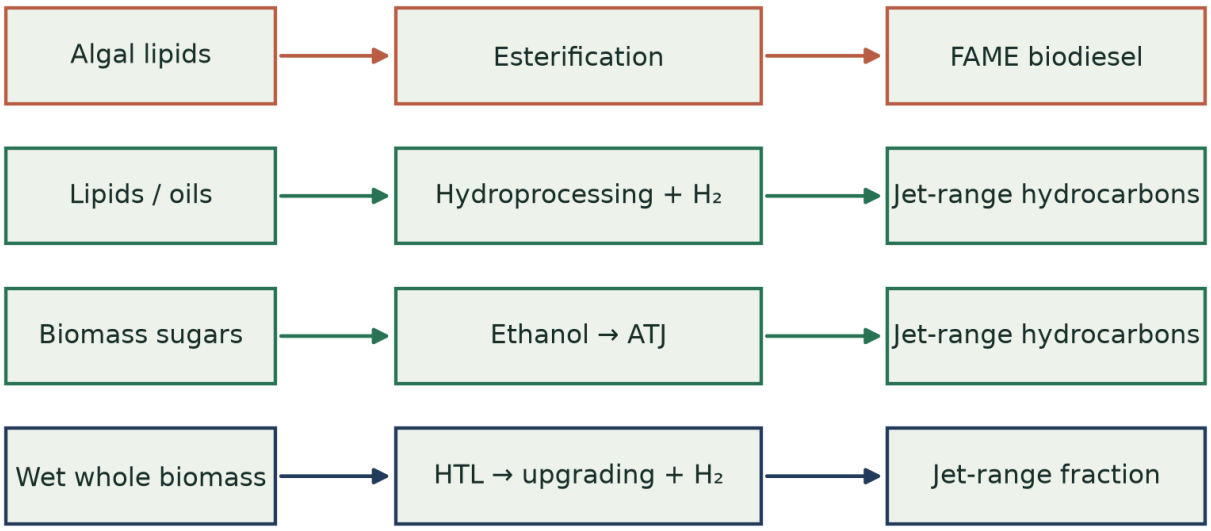
As an illustrative sensitivity calculation, assume 35% useful conversion for fuel and 90% for a battery-electric system. Useful energy per carried kilogram becomes 43.2 × 0.35 = 15.12 MJ versus 0.9 × 0.90 = 0.81 MJ. The ratio is about 18.7, still excluding many design constraints. Those efficiencies are assumed examples, not measured values for a specific aircraft.

Long service lives add inertia

EASA gives a typical passenger-aircraft retirement range of 25–30 years; IRENA describes a usual technical ship lifetime of 25–30 years. These are different metrics, not a forecast that every fleet will turn over in exactly that time. They help explain interest in compatible fuels alongside efficiency improvements and new propulsion systems. [R15, R16]

Sustainable aviation fuel and conversion chemistry

SAF refers to aviation fuels whose technical qualification and sustainability requirements must both be addressed. A fuel that burns in a research test is not automatically approved throughout the fleet. A technically qualified hydrocarbon route also needs a credible life-cycle and feedstock assessment to support a sustainability claim. [R17, R18]



Simplified chemical routes, not a certification map. The HTL row represents a research pathway; upgrading is required. [R17, R20, R21, R23]

Biodiesel usually contains fatty-acid alkyl esters, such as FAME. Hydroprocessing removes oxygen and changes the molecules toward hydrocarbons. In alcohol-to-jet, ethanol can first be dehydrated to ethylene; further chain-building and hydrogenation/upgrading produce suitable hydrocarbon fractions. These steps explain why an algal biodiesel demonstration is not itself a demonstration of SAF.

For an algal starch route, hydrolysis first releases fermentable sugars. Fermentation can produce ethanol, which then enters the separate ATJ conversion. Cultivation, sugar recovery, fermentation and fuel upgrading each introduce yields and energy requirements; an increase in starch content does not determine their combined performance. [R21]

Wet-biomass hydrothermal liquefaction can produce biocrude, followed by hydrotreating and separation. PNNL's 2025 wastewater-algae study reports experiments together with techno-economic modelling; its product-property assessment does not establish certification or routine commercial delivery. [R20]

LanzaJet announced on-spec ethanol-to-jet production at Freedom Pines on 13 November 2025. Use this as a dated company-reported milestone, rather than as evidence of independently audited annual throughput or an algae-based plant. [R19]

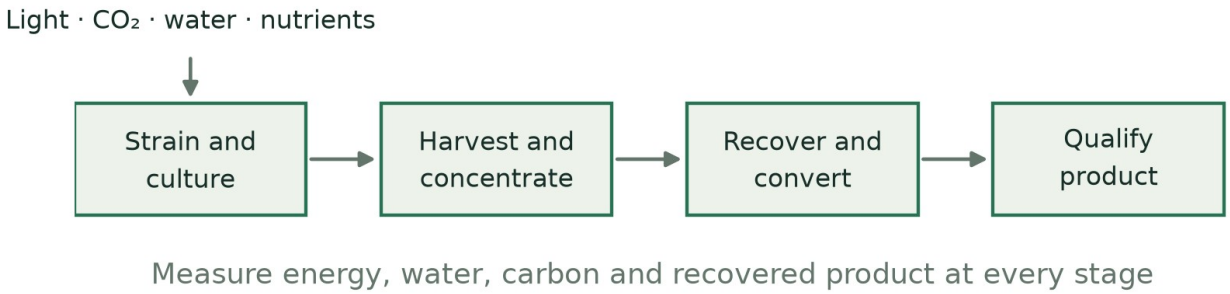
SAF still releases CO_2 during combustion. Its potential benefit arises from a lower assessed life-cycle burden relative to a defined fossil comparator, not zero carbon at the exhaust. Feedstock origin, process energy, hydrogen and land-use effects influence that result. Qualification and blend limits are pathway-specific. [R17, R18]

Algae as organisms and process hosts

“Algae” covers organisms with different cell structures, metabolisms and cultivation requirements. A collage of seaweeds, microalgae and cyanobacteria is useful for showing diversity, but it does not describe one organism. The research discussed here centres on the green microalga Chlamydomonas pacifica. [R24]

What cultivation must supply

A photosynthetic culture needs access to usable light and inorganic carbon as well as water and nutrients. Growth redistributes carbon into several cellular components. Engineering can alter that composition, but a process must still deliver light and nutrients, remove or tolerate heat, circulate the culture and manage contamination. The simplest biological story becomes an engineering problem when these inputs are counted.



A process boundary for asking where energy and material losses occur. Original schematic informed by the research routes. [R20, R23, R24]

What should be measured

Optical density can follow a culture over time, but it needs calibration before it becomes a dry-biomass concentration. Pigments, cell size and aggregation can change an optical signal. Dry cell weight, culture volume, area, time and sampling or harvest history are needed to calculate biomass productivity.

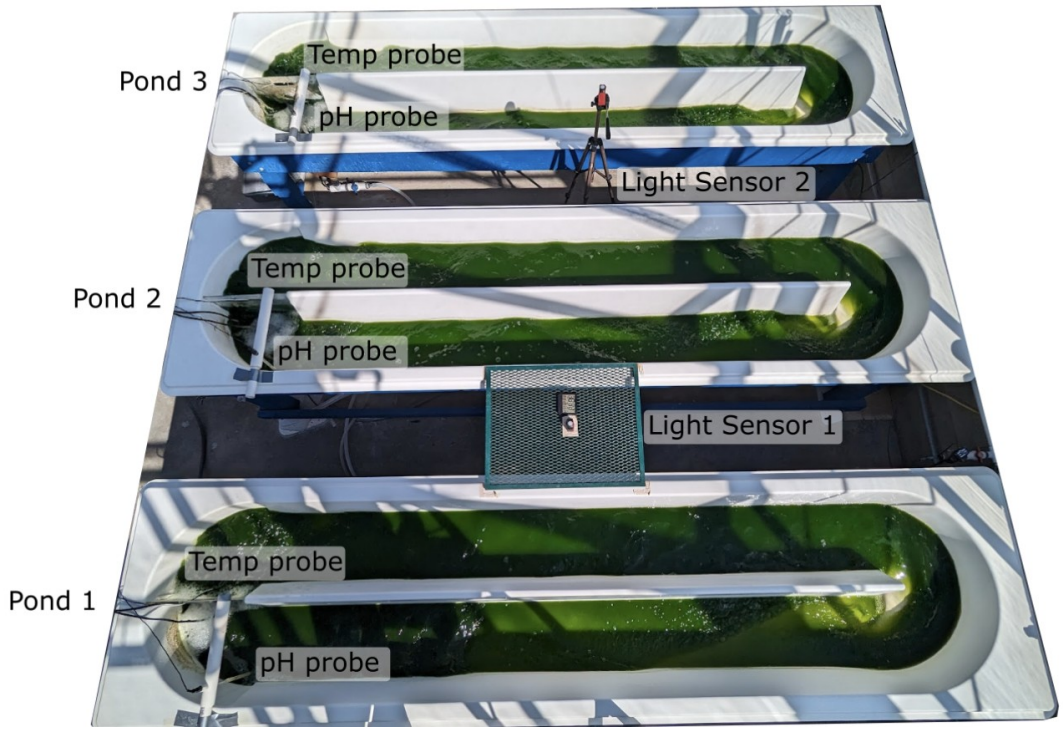
A volumetric biomass concentration in g/L answers a different question from areal productivity in g/m²/day. A percentage of lipid in the dry biomass answers another question again. Recovery and conversion then determine how much useful product leaves the process. These measurements should remain separate until a consistent mass balance links them.

Why host selection matters

The best-known laboratory organism is not automatically the best outdoor production host. Genetic tools, growth stability, product secretion, tolerance and downstream processing all influence the choice. A strain selected for one product can have different requirements from a strain selected for fuel precursors. The next research examples illustrate those choices rather than establishing one universally best alga.

Choosing and evaluating Chlamydomonas pacifica

The species paper describes C. pacifica from a San Diego soil sample, with tolerance tests covering high pH, temperature and salinity. Its abstract reports pH 6–11.5, temperatures up to 42°C and up to 2% NaCl. These are assay-specific observations, not a guarantee of simultaneous optimum production at every extreme. [R24]



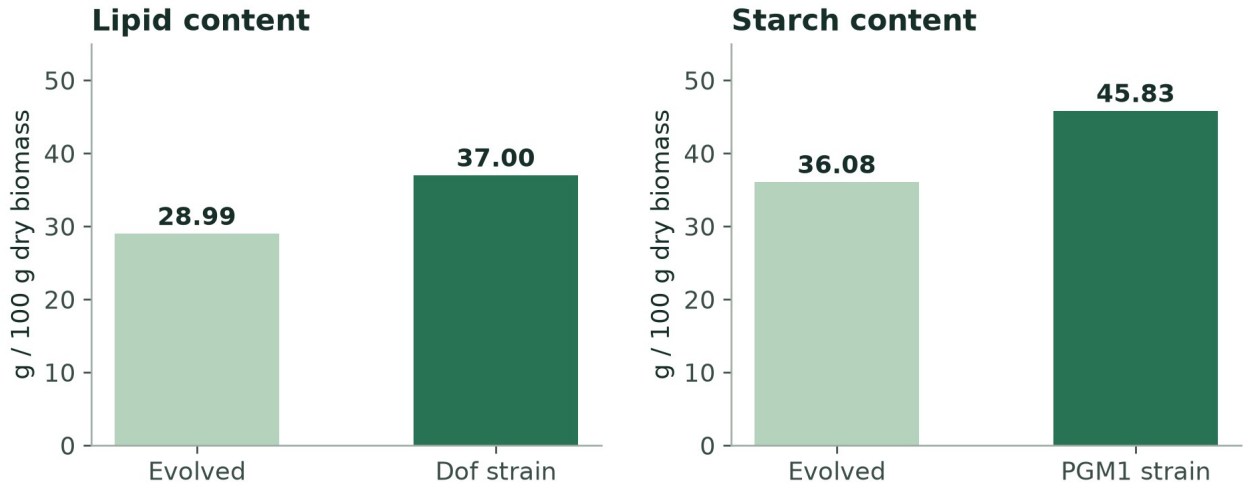
Raceway photograph supplied in the added research deck, slides 6–7. Temperature, pH and light instrumentation illustrate the measurements needed. The associated enzyme-production study is a preprint. [R27]

High tolerance can widen an operating window, but tolerance and productivity are different endpoints. A cell may survive a stressful condition while producing little desired product. For an outdoor process, ask how often that condition occurs, whether the culture recovers, and whether the product phenotype remains stable.

The ACS study combined high-light selection with engineering for lipid or starch accumulation. Pilot cultivation helped test whether a useful phenotype remained observable outside small laboratory cultures. Three well-instrumented ponds can provide an informative experiment, but they do not establish annual commercial reliability. [R23]

The photograph above belongs to the separate enzyme-production work. Its preprint abstract reports three 80-L raceways. Keep the experiment identity attached to each image rather than treating all pond photographs as measurements from the lipid-and-starch study. [R27]

Interpreting the lipid and starch research



Reported central values from the ACS study, Fig. 4e–f. This teaching redraw omits error bars; use the original figure for the authors' uncertainty and replicate definitions. The engineered strains are separate. [R23]

The Dof result concerns regulation associated with lipid accumulation. PGM1 concerns phosphoglucomutase, which interconverts glucose-1-phosphate and glucose-6-phosphate. These strategies connect gene expression and central metabolism to the composition of biomass. They should not be combined into a claim that one strain has all the illustrated gains.

$$\text{Lipid relative gain} = \frac{37.00 - 28.99}{28.99} \times 100 = 27.63\%$$

$$\text{Starch relative gain} = \frac{45.83 - 36.08}{36.08} \times 100 = 27.02\%$$

The lipid increase is 8.01 percentage points of dry biomass; the starch increase is 9.75 percentage points. "About 28% more lipid" describes a relative increase over the reference content. It does not mean an extra 28 percentage points, and it does not by itself mean 28% more fuel per area or time.

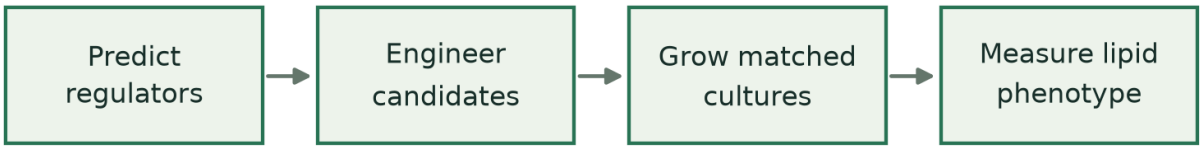
What the experiment establishes

The study demonstrates engineered composition and downstream FAME biodiesel and TPU applications. Reported maximum biomass productivities were 3.90, 3.38 and 2.83 g/m²/day for evolved, Dof and PGM1 cultures, respectively. These maxima cannot be combined with composition measurements from another experiment to calculate a validated production rate. [R23]

The appropriate next engineering question is how much recovered product can be produced per area, time, cost and environmental burden. Answering it requires simultaneous measurements of biomass output, composition and conversion losses, with suitable replication.

From predicted regulators to measured lipid phenotypes

A transcription factor changes gene expression by participating in regulation. Altering a regulator can affect several pathways, so a plausible regulatory target is a hypothesis that needs experimental testing. It is not a direct measurement of metabolic flux.



The logic of the 2025 endogenous-transcription-factor study. Prediction, engineered perturbation and phenotype are separate evidence steps. [R25]

The study predicted 162 TFs computationally, then examined selected candidates. A homology or domain assignment supports a candidate identity; it does not validate every predicted protein's function. Tested strains and controls provide the next evidence layer. [R25]

What a cytometry histogram measures

Nile Red staining was measured in the PE-A channel. Figure 5 shows curves for three biological replicates and a threshold of 7.7×10^4 relative fluorescence units. The horizontal axis is fluorescence, and the distribution describes measured cells. A shifted distribution can support a lipid phenotype under that assay, but it is not a calibrated fuel titer. [R25]

Thousands of measured cells improve description of a population, but they do not create thousands of independent cultures. Biological replicates address culture-to-culture variation. Technical replication addresses variation introduced by a measurement procedure. The sampling hierarchy matters when interpreting significance.

Why use a second measurement

Confocal microscopy locates lipid-associated signal relative to cellular structures. A separate semi-quantitative lipidomic comparison reported TAG relative abundance of 152 ± 21 versus 369 ± 70 per mg biomass for the examined case, a ratio of about 2.43. The unit is relative abundance, not an absolute mass yield or volume of fuel. [R25]

Combining complementary methods can strengthen an interpretation. It still leaves questions about long-term stability, outdoor growth, recovery and process output. A useful paper discussion asks which link in that chain each figure actually tests.

Biobased materials and plastic degrading enzymes

Biobased describes carbon origin. Biodegradable describes biological breakdown under specified conditions. Compostable adds requirements tied to a composting environment. These properties need separate evidence: a biobased polymer can persist, and a biodegradable material may require conditions absent from its disposal environment. [R22]

Polyurethane synthesis joins polyols with diisocyanates. An algal route can supply building blocks for that chemistry, followed by polymer synthesis and performance testing. The ACS study's A2141-coated fabric illustrates a downstream material application; it does not establish every environmental property from the origin of the precursor alone. [R23]

Gunawan and colleagues tested renewable polyester-PU foams for footwear-related properties and degradation. After 12 weeks, the tested samples lost 30% of their mass in compost and 71% in soil. Mass loss and loss of mechanical integrity are informative, but they do not directly quantify complete mineralization of all polymer carbon to CO_2. [R29]

Producing an enzyme is another algal application



Culture growth and plastic treatment are different process operations

The reported PET assay separated enzyme production from a heated reaction. The study also used polyester-PU dispersion assays, which are different substrates and measurements. [R26]

C. reinhardtii was engineered to secrete PHL7. Clearing of Impranil and zymogram activity support polyester-hydrolysing activity. The separate PET experiment used concentrated supernatant at 70°C; TPA was detected in both engineered and control samples, with a stronger signal in the engineered case. The whole-culture PU-film demonstration was illustrative and lacked replicate experiments. [R26]

These results support further development of enzyme production and plastic treatment. They do not demonstrate complete ambient-temperature PET recycling by a growing outdoor culture. Nor do they establish the economics or life-cycle advantage of a commercial recycling plant.

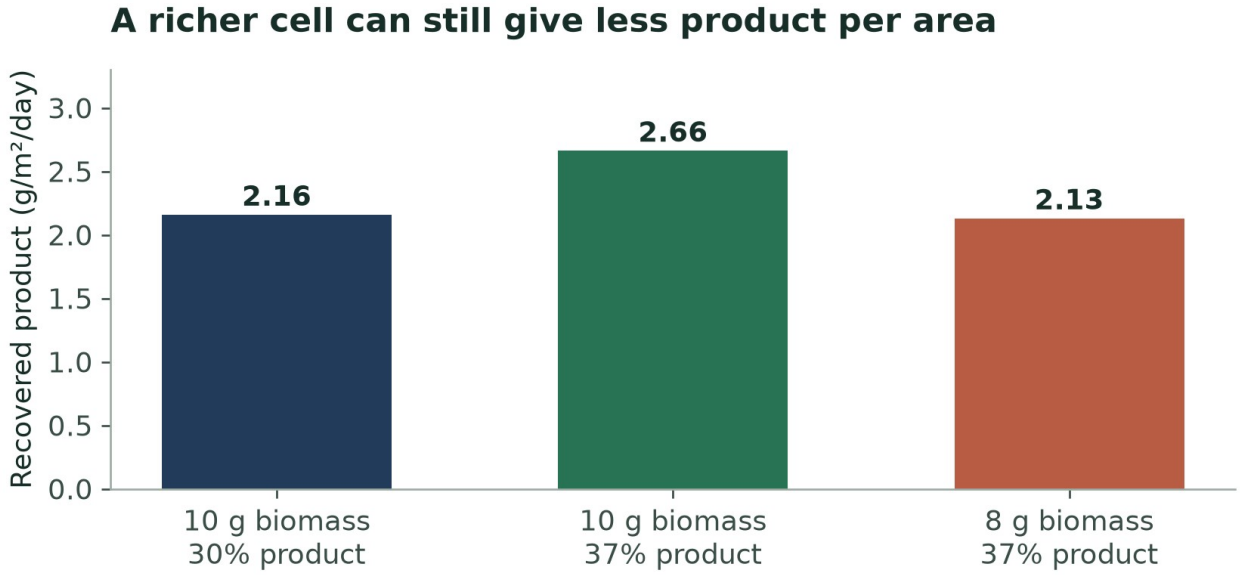
Chemistry guides the claim

Polymer identity matters because enzymes recognize and hydrolyse particular chemical bonds. A result on a polyester-containing PU cannot establish the same degradation rate for PET, another PU formulation or an unrelated polymer. State the substrate, temperature, time, controls and analytical endpoint whenever discussing plastic degradation.

Connecting biomass composition to process output

A process must turn a biological phenotype into recovered material. The following mass-balance framework keeps the relevant terms visible. It is a teaching model, not a fitted representation of the experiments.

$$\text{Recovered product per area and time} = \text{biomass productivity} \times \text{fraction} \times \text{recovery} \times \text{yield}$$



Illustrative inputs only. Recovery is 80% and conversion gives 0.90 g product per g recovered precursor in all three cases. These are not reported experimental values.

At 10 g dry biomass/m²/day, a 0.30 precursor fraction, 80% recovery and 0.90 conversion factor give 2.16 g product/m²/day. Raising the fraction to 0.37 while holding the other terms fixed gives 2.664 g/m²/day. If biomass productivity instead falls to 8 g/m²/day, the output becomes 2.1312 g/m²/day, slightly below the original case.

The product fraction rises by 23.3% in this invented example, yet the final production rate can fall. This is why a richer cell and a better production process are separate claims. A sensitivity calculation identifies measurements that matter; it does not replace simultaneous process data.

Benchmarks need compatible definitions

An NREL state-of-technology report gives an annual average of 16.7 g ash-free dry biomass/m²/day with seasonal cultivation of different strains. That annual, ash-free, rotating-crop benchmark is not directly comparable to a single-strain daily maximum expressed as total dry weight. Match time period, biomass basis, species and harvest accounting before ranking numbers. [R36]

For sustained production, record what leaves the process across the full period. Brief maxima can be useful for diagnosis while annual averages reveal downtime, seasonal effects and operational losses.

Economics and life cycle questions for an algal route

A technically successful product can still require too much energy, money or material. A techno-economic assessment links process assumptions to cost. A life-cycle assessment links them to environmental burdens. Both require a clear system boundary, product specification and treatment of coproducts. [R18, R20]

Choose the service before the comparison

For an aviation fuel, an energy-based unit such as 1 MJ of qualified fuel can help compare routes. For a coating, specify a protected surface area, performance and service life. Comparing equal masses without those functions can favour a material that must be used in a larger amount or replaced more frequently.

Process question	Measurement or assumption needed
Cultivation	Productivity over a stated period; nutrients; CO_2 delivery; mixing
Recovery	Harvest concentration; water removal; solvent or enzyme use
Conversion	Recovered yield; heat; electricity; hydrogen source
Products	Purity; fuel or material performance; usable coproduct quantities
Use and end of life	Combustion or disposal pathway; product lifetime; degradation conditions

A cost estimate with a defined basis

PNNL's 2025 wastewater-algae case reports a modelled minimum fuel selling price of $9.04 per gasoline gallon equivalent, with sensitivity cases of $6.51–13.07/GGE. The energy-equivalent unit is not a physical gallon of jet fuel, and the estimate is not a retail price. Scale, capital cost, yield and coproduct revenues influence the result. [R20]

Coproduct credits require an explicit allocation or displacement method. It is not valid to assume that every byproduct earns unlimited revenue or offsets the full burden of the process. Likewise, renewable feedstock does not remove electricity, hydrogen or nutrient emissions from the boundary.

Read a reduction claim carefully

In an illustrative comparison, a route at 45 g CO_2e/MJ against a 90-g baseline has a 50% reduction: (90 − 45)/90. Those values are invented for the arithmetic. A real SAF reduction requires the applicable life-cycle method and fossil comparator. Lower tailpipe CO_2 alone is not the calculation. [R18]

Using Algae Farm as an engineering exercise

The class ends with Algae Farm, a teaching simulation available in Portuguese. The game makes resource and production decisions visible. Its scores and conversion rules are designed for learning; they are not validated estimates of real fuel certification, commercial yields or life-cycle emissions. [R37]

Record the decision before judging the result

Observation	What to record
Starting conditions	Scenario, organism or contract where shown, starting resources
Decision	The specific setting or action changed
Output	Useful product and any reported losses
Resources	Which resources the route used or exhausted
Interpretation	Whether the result supports the original hypothesis

Compare runs under matched starting conditions whenever possible. Changing several decisions at once can reveal a promising strategy but makes it difficult to identify the cause. A run with a different contract or objective may use a different scoring basis, so raw scores should not automatically be ranked as the same service.

Three debrief questions

Which constraint limited your result? Which additional measurement would let you check your explanation? Would the route still appear preferable if energy, recovery losses and end of life were counted? These questions connect the activity to the research figures and the earlier life-cycle discussion.

A high game score is a hypothesis about a useful process strategy. To test it experimentally, specify an organism, operating conditions, replicated cultures, recovered output and a consistent comparison boundary.

Class links

Presentation: https://molino-algae.github.io/algae-sustainable-future/

Game: https://molino-algae.github.io/justlab-site/games/algae-farm/

Use the class presentation’s live QR code for the active commuting poll. A saved paper QR for an old room can point to an earlier collection, so the instructor should confirm the current session before students answer.

Practice problems

Attempt these without the answer page. Show units and assumptions, not just the final number. Numerical inputs below are teaching scenarios unless a reference is specified.

1 Mixed outward and return journeys

A student reports ten 6-km journeys: six by bus and four by bicycle. Using 0.12242 and 0.00017 kg CO_2e/passenger-km, calculate weekly distance, weekly estimate and mean per journey.

2 Allocation and class totals

Two students share every one of their ten 6-km journeys in the same car, with no other occupants. Use 0.1697080247 kg CO_2e/vehicle-km. Calculate each person's estimate, their combined estimate, passenger-km and unique vehicle-km.

3 A flight distance equivalent

Apply the 0.225-kg short-flight factor to the first student's weekly passenger-distance. Calculate the ratio to that student's commuting estimate. Explain why it is not a literal short flight of the same distance.

4 Relative and absolute change

Lipid content changes from 28.99 to 37.00 g/100 g dry biomass. Calculate the percentage-point increase and relative percentage increase. Explain why neither number alone gives fuel productivity. [R23]

5 A process mass balance

Biomass productivity is 12 g/m²/day, precursor fraction 0.35, recovery 0.75 and conversion 0.90 g product/g recovered precursor. Calculate recovered product per m²/day. How much does a 10% relative loss of biomass productivity change it if other terms stay fixed?

6 Interpreting a cell assay

A Nile Red distribution shifts upward in three engineered cultures relative to three controls. Which conclusion is supported: a changed fluorescence phenotype under the assay, an exact doubled lipid mass, or certified doubled biodiesel output? What extra measurement would help?

7 Energy and climate boundaries

A source reports renewable electricity share of 86.8% in Brazil. Can that be used as a life-cycle carbon factor or as the renewable share of all Brazilian energy? Explain. [R11]

8 An environmental claim

A biobased foam loses mass in a soil test. List the additional evidence needed to support claims about complete mineralization, industrial compostability and lower life-cycle impact than a fossil alternative.

Worked answers

1 Mixed journeys

Distance = 10 × 6 = 60 passenger-km. Bus: 36 × 0.12242 = 4.40712 kg. Bicycle: 24 × 0.00017 = 0.00408 kg. Total = 4.41120 kg CO_2e/week. Mean per journey = 0.44112 kg. No extra ×2 is needed because the entries already count one-way journeys.

2 Shared car

Each person covers 60 passenger-km and receives half the vehicle factor: 60 × 0.1697080247/2 = 5.09124 kg. Combined emissions = 10.18248 kg. Combined passenger-distance = 120 passenger-km, but the shared vehicle travelled only 60 vehicle-km. Allocation shares the vehicle burden; it does not erase the trip.

3 Flight comparison

The equivalent is 60 × 0.225 = 13.5 kg CO_2e. Ratio = 13.5/4.41120 = about 3.06. The factor describes a 300–1,000-km flight profile applied to equal passenger-distance; it is not a model of a literal 60-km flight.

4 Composition change

The increase is 37.00 − 28.99 = 8.01 percentage points. Relative change = 8.01/28.99 × 100 = 27.63%. Fuel output additionally needs biomass production, recovered fraction and conversion yield measured on a compatible basis.

5 Product balance

12 × 0.35 × 0.75 × 0.90 = 2.835 g product/m²/day. A 10% relative fall in biomass productivity changes 12 to 10.8 and the output to 2.5515 g/m²/day, also a 10% fall when all other factors remain fixed.

6 Assay interpretation

The fluorescence phenotype is supported under the tested conditions. Calibrated chemical analysis can help quantify lipid mass or composition. Independent culture replication, dry biomass and process conversion measurements are required before inferring fuel productivity.

7 Electricity share

No. It is a share of electricity supply in a defined geography and year. A carbon factor requires emissions per unit of energy with a stated boundary. Total energy includes additional fuels and end uses beyond electricity.

8 Material claims

Mineralization needs a carbon-fate measurement with controls, such as suitable respirometry rather than mass loss alone. Compostability requires evidence under the applicable composting conditions and criteria. A lower-impact claim requires a matched life-cycle comparison with functional performance, lifetime, production inputs and end of life. None follows solely from biobased origin.

Terms and connections to remember

Term	Meaning in this class
Functional unit	The quantified service used to compare alternatives.
Passenger kilometre	One person transported over one kilometre.
Vehicle kilometre	One vehicle travelling one kilometre.
CO_2e	Climate effects expressed with a specified equivalence method.
System boundary	Activities included in the accounting or experiment.
Primary energy	Energy before conversion to secondary carriers; datasets must state the method.
FAME	Fatty-acid methyl esters, a biodiesel product class.
SAF	Aviation fuel requiring technical qualification and a sustainability assessment.
HTL	Hydrothermal liquefaction of wet feedstock to a biocrude intermediate.
Biomass concentration	Biomass mass per culture volume, often g/L.
Biomass productivity	Biomass output per time and volume or area.
Content	A constituent fraction of biomass on a stated mass basis.
Recovery	The fraction of a target precursor recovered downstream.
Conversion yield	Product obtained per input on a defined mass or molar basis.
Transcription factor	A protein involved in regulation of gene expression.
Biological replicate	An independent biological experimental unit.
Technical replicate	A repeated measurement or procedure within an experimental unit.
Biobased	Derived wholly or partly from biological carbon feedstocks.
Biodegradation	Biological breakdown under specified conditions.
TEA and LCA	Economic and environmental assessments linked to an explicit process model.

The common engineering task is to connect these levels without skipping evidence: carbon source, organism, measured phenotype, recovered product, useful service and life-cycle result. Each link needs compatible units and a defined comparison.

References

Reference codes match the Stage 1 evidence audit. Historical scenarios retain their dates. Original research and official methods are linked below. Full source metadata accompanies this package.

R01 Supplied Sustainable Future course: source index

Supplied Sustainable Future teaching collection. Lecture 1, PDF p. 9 provides the historical 2022 world energy snapshot. Other relevant figures are indexed in the Stage 1 audit.

R02 OWID: Energy mix

Methods; August 2026 revision. Historical slide snapshot retained separately.

Read the original source

R03 OWID: Fossil fuels

Global fossil fuel consumption chart; Energy Institute and Smil underlying data.

Read the original source

R04 OWID: CO_2 emissions from transport

2018 modal breakdown; underlying IEA / ICCT.

Read the original source

R05 ADEME / Impact CO_2: frozen transport data

Archived source file; downloaded and checked against calculator constants.

Read the original source

R06 ADEME: gasoline compact/medium car

170 g CO_2e per passenger-km displayed, rounded; manufacturing and fuel breakdown.

Read the original source

R07 ADEME: thermal bus

122 g CO_2e per passenger-km displayed, rounded.

Read the original source

R08 Blondel, Mispelon & Ferguson (2011), Cycle more Often 2 cool down the planet!

Printed pp. 9–11; bicycle lifetime, manufacture/maintenance and dietary assumptions.

Read the original source

R09 ICCT (2025): Life-cycle GHG emissions of passenger cars in the EU

Summary comparison: medium cars sold in 2025; lifetime electricity 2025–2044.

Read the original source

References continued

R10 Mera et al. / ICCT (2023): Passenger-car life-cycle GHG emissions in Brazil

Printed pp. 15–17, Figs. 5–7; 22-year life, 288,000 km; electricity/fuels 2023–2044.

Read the original source

R11 EPE (2026): BEN 2026, base year 2025

Printed p. 35: 86.8% renewable share of domestic electricity supply (OIEE), including net imports.

Read the original source

R13 Boeing CASCADE documentation: energy and cost parameters

Table 58: Jet A lower heating value 43.2 MJ/kg; kWh-to-MJ conversion.

Read the original source

R14 Embry-Riddle: Electric aircraft

Battery pack specific energy about 0.20–0.25 kWh/kg; distinguish pack and cell.

Read the original source

R15 EASA: Sustainability in the aircraft end-of-life phase

Passenger aircraft typically retired after 25–30 years; cargo often 30–40.

Read the original source

R16 IRENA (2021): A pathway to decarbonise the shipping sector by 2050

Printed p. 25: technical ship lifetime usually 25–30 years.

Read the original source

R17 ICAO: SAF conversion processes

Approved Conversion Processes table; narrative explicitly dated July 2023.

Read the original source

R18 ICAO: CORSIA fuels life-cycle emissions

Fuel life-cycle methodology, including core emissions and land-use effects.

Read the original source

R19 LanzaJet (13 November 2025): Freedom Pines on-spec fuel announcement

Company announcement of on-spec jet fuel and full plant operation.

Read the original source

References continued

R20 Kumar et al. (2025): HTL of wastewater-grown algae; PNNL record

Energy Conversion and Management: X 27, 101096. DOI: 10.1016/j.ecmx.2025.101096.

Read the original source

R21 Zhou & Hu (2020): Catalytic Thermochemical Conversion of Algae and Upgrading of Algal Oil for the Production of High-Grade Liquid Fuel: A Review

Catalysts 10(2), 145; DOI: 10.3390/catal10020145. Conceptual conversion diagram, not proof of this strain's fuel yield.

Read the original source

R22 European Commission: biobased, biodegradable and compostable plastics

Definitions distinguish feedstock origin from biodegradation and composting conditions.

Read the original source

R23 Gupta & Dutra Molino et al. (2025): Engineering C. pacifica for high lipid and starch

ACS ES&T Engineering 5 (2025), 36–49. First online 25 November 2024. DOI 10.1021/acsestengg.4c00443. See Figs. 1 and 4–6.

Read the original source

R24 Dutra Molino et al. (2025): Description of C. pacifica

Algal Research 88 (2025), 104034. DOI 10.1016/j.algal.2025.104034. Species characterization and tolerance assays.

Read the original source

R25 Gupta, Oliver, Dutra Molino et al. (2025): endogenous transcription factors

Algal Research 90 (2025), 104192. DOI 10.1016/j.algal.2025.104192. See Figs. 5–6, assay methods and lipidomics.

Read the original source

R26 Dutra Molino et al. (2025): PHL7 secretion from C. reinhardtii

Scientific Reports 15 (2025), 24690. DOI 10.1038/s41598-025-09100-0. Substrate-specific enzyme and polymer assays.

Read the original source

R27 Diaz & Dutra Molino et al. (2025): PHL7 production in C. pacifica

bioRxiv preprint v1, 15 May 2025. DOI 10.1101/2025.05.14.654053. Three 80-L raceways described in the abstract; the separate numerical pond maxima remain unverified in the audit.

Read the original source

R29 Gunawan et al. (2020): biodegradation of renewable PU foams

Bioresource Technology Reports 11 (2020), 100513. DOI 10.1016/j.biteb.2020.100513. Formulation-specific soil and compost tests.

Read the original source

References continued

R32 ADEME: short flight

225 g CO_2e/passenger-km; 300–1,000 km flights; includes contrail effects, excludes infrastructure.

Read the original source

R33 ADEME: medium flight

185 g CO_2e/passenger-km; 1,000–2,000 km flights.

Read the original source

R34 ADEME: long flight

178 g CO_2e/passenger-km; flights over 5,000 km.

Read the original source

R35 UK government (2026): GHG conversion-factor methodology

Alternative travel factors; separate cabin, distance, upstream emissions and non-CO_2 treatment.

Read the original source

R36 Klein, Davis & Wiatrowski (2024): Open-pond algal biomass state of technology

NREL/TP-5100-88802, DOI: 10.2172/2370102; 2023 annual average 16.7 g AFDW/m^2/day, seasonal crop rotation.

Read the original source

R37 Algae Farm: current teaching game

Teaching simulation; resource/score rules are not validated process or LCA coefficients.

Read the original source